

Stigma, Concealment, and Marriage Matching: Biomarker Evidence from Korea^{*}

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Abstract

Social norms and stigma are central to many economic outcomes but rarely observed directly. This paper provides a behavioural measure of stigma in a setting where the underlying trait can be biologically verified. Using twelve waves of the Korea National Health and Nutrition Examination Survey (2008–2021), I compare self-reported smoking against biomarker-verified smoking measured by urinary cotinine concentration for 8,768 married couples and 7,940 single adults. I document a 34 percentage point gendered concealment gap where biomarker-positive women are far more likely than biomarker-positive men to self-report as nonsmokers, with married women showing an additional increment. Within the biomarker-positive sample of wives, concealment is systematically associated with husband education, surviving controls for the wife's own education and age. The choice of measurement affects empirical conclusions about marital sorting. The wife smoking–husband education relationship is 39 percent larger in magnitude under self-report than under biomarker measurement, and structural Choo-Siow surplus estimates differ systematically across measurement schemes. The gendered concealment gap has declined substantially over the study period.

JEL codes: D10; J12; J16.

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1 Introduction

Social norms and stigma are central to many economic outcomes, including labour-market behaviour, marriage matching, political participation, and the take-up of public programs, but they are rarely observed directly. A social norm describes a shared expectation about appropriate behaviour; stigma arises when violating such a norm attaches social disapproval, shame, or loss of status to a person. One implication of stigma is concealment. Individuals who possess a stigmatized trait may strategically hide it, especially when disclosure is costly for social relationships, marriage-market prospects, or public reputation.

Measuring stigma in specific social contexts matters because it shapes economically consequential decisions (Görges and Nosenzo, 2020). Existing economic research has measured norms and stigma in several ways. Some studies ask the beliefs and elicit perceived norms directly, for example using incentivized coordination games that ask respondents to identify what others consider socially appropriate (Camerer *et al.*, 2004; Charness *et al.*, 2025; Krupka and Weber, 2013; Bursztyn *et al.*, 2020; Bicchieri and Xiao, 2009). Those methods can efficiently elicit the first- and second-order beliefs, however they cannot test to what extent individuals' willingness to follow those stated norms (Görges and Nosenzo, 2020). The another strand of literature infers norms from actual behaviours to tackle this problem. Some related literature study stigma through survey misreporting, comparing self-reported outcomes with administrative records to identify concealment of socially sensitive statuses such as welfare-program participation (Celhay *et al.*, 2024, 2025). Above method implies that although norms and stigma are latent, they can be inferred from the behaviours they induce.

This paper provides such a behavioural measure of stigma in a setting where the stigmatized trait can be biologically verified. I study Korean female smoking, a behaviour subject to strong gender-asymmetric stigma deeply rooted in Confucian gender norms (Park and Cho, 1995; Woo, 2018). Using twelve waves of the Korea National Health and Nutrition Examination Survey (2008–2021), I compare self-reported smoking against biomarker-verified smoking measured by urinary cotinine concentration for 8,768 married couples and 7,940 single adults. The biomarker provides objective ground truth that is unaffected by reporting incentives, allowing us to identify directly the share of biological smokers who self-report as nonsmokers, which is a behavioural measure of concealment that responds to underlying social cost.

The biomarker approach has two advantages. First, it provides an objective benchmark for the underlying behaviour, allowing smoking status to be separated from willingness to disclose smoking. Second, it allows concealment to be measured at the individual level and compared across

gender, marital status, spouse characteristics, and birth cohorts. I use this measure to document a gendered concealment gap, examine whether concealment among wives is related to husband characteristics, and show how self-reported smoking can distort estimates of marriage matching. The paper therefore links the measurement of stigma-induced concealment to a substantive economic application: assortative mating on smoking and education.

I document four findings. First, a large gendered concealment gap. Among biomarker-positive adults, women are 34 percentage points more likely than men to self-report as nonsmokers, with married women showing an additional 10 percentage point increment relative to single women. Among biomarker-positive wives, approximately half (46 percent) report themselves as nonsmokers; the corresponding rate among biomarker-positive husbands is only 11 percent. Second, concealment is not random. Within the biomarker-positive sample of married women, those married to higher-educated husbands are systematically more likely to conceal, and the pattern survives controls for the wife's own education and demographic characteristics. Concealers' husbands lie between those of true nonsmokers and those of disclosers on multiple matching outcomes, consistent with concealment being correlated with partner status. Third, the choice of measurement substantially affects empirical conclusions about marital sorting on smoking. The relationship between wife smoking and husband education is roughly 39 percent larger in magnitude when smoking is measured by self-report than when measured by biomarker. Smoking homogamy estimates, in contrast, are robust across measurement schemes because both spouses' smoking is mismeasured in parallel. Fourth, the gendered concealment gap has fallen substantially over time, from approximately 71 percent of biomarker-positive wives in 2008 to 46 percent in 2021, with both within-cohort norm change and cohort replacement dynamics.

First, this paper contribute to a growing literature on measuring stigma and social norms through observable behaviour. One strand of this literature elicits perceived norms directly through surveys, incentivized coordination games and belief-elicitation experiments ([Krupka and Weber, 2013](#); [Camerer *et al.*, 2004](#); [Kimbrough and Vostroknutov, 2018](#); [Charness *et al.*, 2025](#)), recovering subjects' beliefs about what others consider socially appropriate ([Bursztyn *et al.*, 2020](#)). A complementary strand infers the size and structure of social norms from observed economic behaviour: stigma in welfare-program take-up ([Moffitt, 1983](#); [Besley and Coate, 1992](#); [Celhay *et al.*, 2024, 2025](#)), social pressure in voting ([DellaVigna *et al.*, 2016](#); [Funk, 2010](#)), well-being consequences of social conformity ([Clark, 2003](#)), and the broader influence of social image on economic choice ([Bursztyn and Jensen, 2017](#)). The strand most directly related to ours uses linked survey-to-administrative data to identify misreporting of stigmatized behaviours, with [Celhay *et al.* \(2024\)](#) documenting that respondents in high-stigma neighbourhoods are more likely to under-report welfare receipt and

[Celhay *et al.* \(2025\)](#) showing similar patterns across multiple program domains. I complement this work by introducing a biomarker-based approach to behavioural validation which could directly observe the true individual status. In terms of the large gender gap of misreporting behaviours, I could infer the existence of stigma and track the time trends of its revolution.

Second, this paper contribute to an emerging literature on gender concealment gaps. [Exley *et al.* \(2024\)](#) study U.S. university students given the option to conceal poor academic grades and find that male students conceal more than female students, with inference-based reputational costs as the proposed mechanism: women may refrain from concealing because employers infer worse hidden grades for female concealers than for male concealers. I study a setting where the direction reverses and find a substantially larger gender concealment gap (34 percentage points among biomarker-positive adults). Our findings establish that the existence and direction of a gender concealment gap depends on which gender faces social cost for the trait being concealed. The Korean setting also allows us to measure concealment behaviour against biological ground truth, complementing the experimental and quasi-experimental designs typical in this literature. More broadly, our results connect to a body of work documenting gender differences in self-promotion and information disclosure in labour markets and educational settings ([Coffman, 2014](#); [Exley and Kessler, 2022](#); [Murciano-Goroff, 2022](#)), though the specific mechanisms (norm-based stigma versus inference-based reputation costs) differ across contexts.

Third, this paper contribute to the empirical literature on bidimensional marriage matching. The foundational structural framework of [Choo and Siow \(2006\)](#) has been applied to a range of marital sorting dimensions, including BMI ([Oreffice and Quintana-Domeque, 2010](#)), BMI and education ([Chiappori *et al.*, 2012](#)), race and physical characteristics ([Chiappori *et al.*, 2016](#)), and smoking alongside education ([Chiappori *et al.*, 2018](#)). A common assumption across this literature is that the matching traits of interest are observed correctly. We show that this assumption matters empirically. When one matching trait is subject to gendered stigma and is systematically misreported by one side of the market, both reduced-form matching coefficients and structural surplus estimates differ substantively between self-report and biomarker-verified measurement. To my knowledge, this is the first to apply biomarker validation to the matching-trait measurement problem in this literature, and our results suggest that matching estimates on potentially stigmatized traits which include not only smoking but also drug use, sexual behaviour, and politically sensitive attitudes should be interpreted with care in settings where the trait in question is subject to differential disclosure by sex or social group.

The remainder of the paper is organized as follows. Section 2 describes the Korean institutional context, including the cultural origins of female smoking stigma and the structure of the

Korean marriage market. Section 3 describes the KNHANES data and our analytic samples. Section 4 sets out our empirical strategy. Section 5 presents the main results and robustness analyses. Section 6 concludes.

2 Background and Context

This section provides the substantive context for our analysis. I first document the well-known divergence between self-reported and biomarker-verified female smoking in Korea and East Asia more broadly (Section 2.1), then trace the cultural origins of the gendered stigma in Confucian gender norms (Section 2.2). Section 2.3 discusses the Korean marriage market and the role of education-based assortative matching.

2.1 Evidence of gendered smoking concealment in Korea

A robust finding in the Korean public health literature is that female smoking prevalence is substantially under-reported in survey self-reports compared with biomarker-verified measures. [Park *et al.* \(2014\)](#) compare self-reported smoking against urinary cotinine concentrations and document a striking discrepancy: based on self-reports, the current smoking rate was 7.1% among women and 47.8% among men, while cotinine-verified rates were 18.2% and 55.1% respectively. The biomarker thus reveals that roughly 60% of biologically smoking women self-reported as non-smokers, compared to only 13% of biologically smoking men. Similar findings are also explored in [Jung-Choi *et al.* \(2011\)](#) where they used earlier wave of the same dataset.

These findings establish a baseline empirical fact: Korean female smokers under-report their smoking behaviour at much higher rates than Korean male smokers. The natural inference is that disclosure is not a passive measurement but an active behavioural response to a social environment that penalizes one gender more than the other. As a consequence, self-reported female smoking identifies a selected subset of biological smokers who are those willing to disclose rather than the full population of biologically smoking women.

2.2 Cultural origins: Confucian gender norms

The gendered structure of smoking stigma in Korea has deep historical roots in the Confucian value system that has shaped Korean (along with other Eastern Asian population) family and social organization for centuries ([Park and Cho, 1995](#)). Confucianism places importance on the family unit and on harmonious human relationships, with harmony understood to require that each individual fulfil prescribed roles within a hierarchical social structure. Traditionally, this frame-

work assigned women responsibility for managing household affairs and raising children, while men were positioned as primary breadwinners and representatives of the family in public life. This division has supported a patriarchal family structure with substantial intra-household gender inequality, particularly in bargaining power, behavioural autonomy, and female inheritance (Ho *et al.*, 2004).

These values have shaped social attitudes toward smoking in gender-specific ways. Male smoking is widely tolerated and in some professional contexts even expected as part of male social bonding rituals. Female smoking, in contrast, is broadly perceived as conflicting with traditional expectations of women as wives and mothers (Woo, 2018). A Korean woman who smokes may face social criticism both for adopting what is seen as a predominantly male behaviour and for visibly departing from the cultural ideal of feminine norms.

The consequence is that Korean women face substantially stronger social penalties for smoking than Korean men, creating an incentive structure that favours concealment (Woo, 2018). While I do not directly observe penalties in specific domains, the gendered concealment I document is consistent with women experiencing stigma costs across multiple settings, including professional environments, family relationships, and marriage market.

2.3 Korean marriage market and matching on education

Korea has historically exhibited strong educational homogamy in marriage, comparable to or exceeding patterns observed in other East Asian and Western contexts (Smits and Park, 2009). Education-based assortative matching has remained stable or strengthened in recent decades despite rapid expansion of female educational attainment, and serves as a primary mechanism by which inter-generational socioeconomic position is transmitted in Korean society.

I focus on smoking as an additional sorting dimension alongside education, following the bidimensional matching framework of Chiappori *et al.* (2018). While the marriage-market consequences of smoking have been studied in U.S. context, to my knowledge no prior work has examined smoking-based matching in Korea or in any setting where the trait being studied is subject to a sharp gendered stigma. The combination of strong gender-asymmetric smoking norms, strong educational assortativeness, and the availability of biomarker validation makes the Korean setting particularly well-suited to study how stigma-driven misreporting interacts with marital matching estimates.

3 Data

3.1 Data Source

I use twelve waves of the Korea National Health and Nutrition Examination Survey (KNHANES), a nationally representative, cross-sectional health survey conducted annually by the Korea Disease Control and Prevention Agency. KNHANES combines a face-to-face health interview, a self-administered nutrition questionnaire, and an in-person health examination. The examination component includes a urine collection from which urinary cotinine concentration is measured for a rotating subsample of respondents. The survey design is a stratified, multi-stage probability sample of households nested within enumeration districts, and is described in detail in [Kweon *et al.* \(2014\)](#).

I restrict attention to waves with available cotinine data: 2008, 2009, 2010, 2011, 2014–2021. The 2012 and 2013 waves did not collect urinary cotinine and are excluded. Cotinine is a major metabolite of nicotine, is widely used as an objective biomarker of recent tobacco exposure ([Benowitz *et al.*, 2009, 2019](#)). Compared with nicotine itself, cotinine has a substantially longer elimination half-life: the CDC’s NHANES laboratory documentation reports an estimated half-life of 15–20 hours for cotinine, compared with 0.5–3 hours for nicotine. Cotinine can be measured in serum, urine, and saliva, and urinary concentrations are typically several times higher than serum concentrations, making urine a useful non-invasive matrix for detecting recent exposure ([Centers for Disease Control and Prevention, 2020](#)). In this study, urinary cotinine therefore provides an objective measure of nicotine exposure in our sample. Cotinine was measured using gas chromatography–mass spectrometry through 2015 and liquid chromatography–mass spectrometry from 2016 onward. Those methods have been validated for measuring cotinine and related nicotine metabolites with high accuracy and are widely used in epidemiological and toxicological studies.

3.2 Sample construction

Our analyses use three samples derived from the pooled KNHANES adult population (age 18 and over).

Adults sample. The base sample consists of all KNHANES respondents aged 18 or older across the twelve waves, with valid sex and marital status. This yields 79,208 adults, of whom 44,569 (56.3%) have a valid urinary cotinine measurement.

Singles sample. A subset of the adults sample restricted to those reporting never-married mar-

ital status (*marri_1* = 2). This sample contains 13,689 single adults, of whom 7,940 have a cotinine measurement.

Couples sample. I link married couples within each household using KNHANES family-relationship variables. For waves 2008–2009, the variable *relat* identifies the survey respondent (code 10) and the respondent’s spouse (code 20). For waves 2010 onward, the analogous codes are *fam_rela* = 1 (respondent) and *fam_rela* = 2 (spouse). Within each household I link the unique respondent and the unique opposite-sex spouse, provided both report being currently married. Households without exactly one such pair are dropped.

I identify 22,778 linked couples in total. Of these, 9,509 have a valid cotinine measurement on both spouses; after additionally requiring valid information on both spouses’ education and age, our analytic working sample contains 8,768 couples.

A nontrivial share of married adults (4,931 individuals) are not captured by this linkage rule because they belong to multi-couple households. Diagnostic analysis indicates these unlinked married adults are systematically older (mean age 61.2, median 66) than the linked respondents (mean 54.7, median 54), consistent with coresident elderly parents or grandparents in multi-generational households who are either married or experiencing bereavement rather than coresident married children. I therefore restrict attention to primary-respondent couples and acknowledge in the discussion that our sample reflects Korean households’ primary respondent couples.

3.3 Smoking measurement

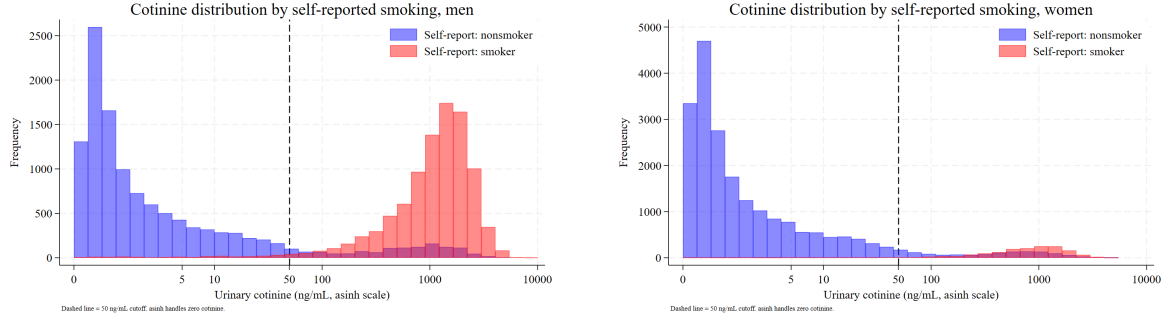
I use two distinct measures of smoking: self-report and biomarker. The self-reported smoker indicator follows the convention in [Chiappori *et al.* \(2018\)](#), hereafter COQ, classifying as a current smoker any respondent who (i) reports having smoked at least 100 cigarettes in their lifetime and (ii) currently smokes either daily or occasionally.¹ Our principal definition harmonizes these by treating both daily and occasional smoking as current in the later waves. For robustness, I also construct a stricter “daily-only” indicator that includes only code 1, which is equivalent to the 2008–2009 definition applied throughout.

The biomarker measure uses urinary cotinine concentration in ng/mL. I classify a respondent as a biological smoker if urinary cotinine is at or above 50 ng/mL, the cutoff established in the Korean validation literature ([Park *et al.*, 2014](#)). [Figure 1a](#) and [Figure 1b](#) show that the cotinine

¹The questionnaire variable for current smoking, *BS3_1*, uses different codings before and after 2010: in 2008–2009 the question distinguishes only “currently smokes” (code 1) from “used to smoke, but no longer” (code 2), whereas from 2010 onward respondents distinguish “smoke daily” (code 1), “smoke occasionally” (code 2), and “used to smoke” (code 3).

distribution is sharply bimodal for both men and women, with the 50 ng/mL cutoff falling in the natural valley between the two modes; fewer than 3.5% of respondents fall in the ambiguous range of 30–100 ng/mL.

Figure 1: Cotinine distribution by self-reported smoking status and gender



(a) Cotinine distribution by self-reported smoking status, men

(b) Cotinine distribution by self-reported smoking status, women

3.4 Three-group classification

For analyses comparing self-report and biomarker measurement, I construct a three-group categorical variable among respondents with valid cotinine measurement:

- *True nonsmoker*: cotinine-negative (<50 ng/mL) and self-reports as nonsmoker.
- *Concealer*: cotinine-positive (≥ 50 ng/mL) but self-reports as nonsmoker.
- *Discloser*: cotinine-positive and self-reports as smoker.

A fourth potential category is cotinine-negative but self-reports as smoker, which is small (typically ex-smokers near the cutoff or the light smokers who did not smoke in the time window of health examination) and is excluded from the three-group comparison.

3.5 Other variables

Education. I use two education variables. The detailed measure converts a nine-category education variable to years of schooling using the mapping: pre-school or less = 0, elementary = 6, middle school = 9, high school = 12, 2–3 year college = 14, four-year college = 16, graduate school = 18. The binary high-education indicator equals one for high school graduate or above.

Drinking. As a placebo health behaviour less stigmatized for Korean women than smoking,

I use a monthly-drinker indicator (drinks at least monthly in the past year. KNHANES does not collect an alcohol biomarker, so I use self-reported drinking throughout.

Region. KNHANES identifies seventeen administrative regions. I construct a metropolitan indicator equal to one for Seoul and the six other metropolitan cities (region codes 1–7) and zero for provinces (codes 8–17), following standard Korean urban classifications.

Birth cohort. Computed as wave year minus age. I define five cohort groups for heterogeneity analysis: pre-1950, 1950–59, 1960–69, 1970–79, and 1980 or later.

Matching outcomes. Within linked couples, I define the education gap as husband’s education years minus wife’s. I construct binary indicators for whether the husband has more education than the wife (“marry up” from the wife’s perspective) and the reverse.

3.6 Summary statistics

Table 1 reports summary statistics for the analytic sample. The married sample is the linked-couples working dataset with cotinine on both spouses (8,822 couples; 17,644 individuals split as 8,822 husbands and 8,822 wives). The singles sample consists of never-married adults with valid cotinine ($N = 4,787$ men and 3,153 women). Means with standard deviations in parentheses are reported for continuous variables; for binary variables only means are reported.

Several features of Table 1 are worth noting. Self-reported smoking is much more common among men than women: married men report current smoking at a rate of approximately 39.1%, married women at approximately 4.3%. Biomarker-positive smoking rates are systematically higher than self-reported rates for both sexes, but the gap is dramatically larger for women: cotinine-positive smoking is approximately 9.6% for married women, nearly double the self-reported rate, while for married men the gap is approximately 3 percentage points. This difference between biomarker and self-report measurement is the central object of interest in our subsequent analysis.

Singles are substantially younger than married respondents (mean age 27–29 for singles versus 51–54 for married) and more educated. This age difference reflects the demographic structure of Korean marriage timing rather than a sample artifact; I account for it through cohort and age controls in our regression analyses.

A full-sample version of Table 1 appears in Appendix Table A.1, which uses the unrestricted adult KNHANES sample for comparison. Differences between the analytic and full samples are modest, consistent with the cotinine subsample being broadly representative of the

Table 1: Summary statistics of analytic sample

	Couples sample		Singles sample	
	Married men	Married women	Single men	Single women
<i>Panel A: Demographics</i>				
Age	54.299 (13.565)	51.057 (13.116)	29.400 (9.951)	26.990 (9.043)
Education (years)	10.177 (4.624)	8.945 (4.871)	12.434 (2.857)	12.970 (2.515)
High education (HS+)	0.696	0.626	0.913	0.941
Body mass index	24.388 (3.079)	23.708 (3.427)	24.154 (4.009)	21.795 (3.691)
<i>Panel B: Smoking</i>				
Lifetime ≥ 100 cigarettes	0.805	0.083	0.650	0.191
Self-report current smoker	0.391	0.043	0.484	0.123
Self-report daily smoker	0.363	0.034	0.431	0.099
Cotinine-positive (≥ 50 ng/mL)	0.423	0.096	0.548	0.195
Concealer (cot+, SR no)	0.097	0.566	0.141	0.408
Cotinine level (ng/mL)	559.475 (862.750)	80.783 (325.566)	720.825 (922.282)	153.489 (415.051)
Cigs/day, current smokers	15.639 (7.967)	8.628 (7.050)	13.912 (22.035)	8.660 (6.035)
Age first smoked	19.757 (4.240)	23.995 (9.280)	18.352 (3.145)	18.860 (4.037)
<i>Panel C: Other behavior & location</i>				
Monthly drinker	0.725	0.396	0.752	0.611
Metropolitan resident	0.427	0.427	0.499	0.548
Observations	8822	8822	4787	3153

Notes: This table reports the summary statistics of our analytic sample. Married men and women are linked spouses in couples with cotinine on both spouses. Single men and women are never-married adults with valid cotinine measurement. Means with standard deviations in parentheses for continuous variables; means only for binary variables. Cigs/day computed among self-reported current smokers only. Current smoker = self-reports daily or occasional smoking with lifetime ≥ 100 cigarettes. Cotinine-positive = urinary cotinine ≥ 50 ng/mL. Concealer rate is among the full sex-marital group (denominator includes nonsmokers).

underlying adult population.

3.7 Sample selection table

Table [Table 2](#) shows the construction of the analytic sample from the raw KNHANES data. Successive restrictions remove adults under 18, observations with missing sex or marital status, and for couples those who cannot be linked into a primary respondent–spouse pair. The final analytic couples sample of 8,822 reflects the additional requirement of valid cotinine and education on both spouses.

Table 2: Sample construction

Step	N	Lost
<i>Panel A: Adults sample (KNHANES respondents pooled)</i>		
All KNHANES respondents, 2008–2011 & 2014–2021	99,511	
After age ≥ 18	79,311	20,200
After valid sex and marital status	79,208	103
<i>Panel B: Linked-couples sample</i>		
Married, linked to opposite-sex spouse	22,778	
Both spouses have cotinine measurement	9,509	13,269
Both spouses have valid education and age	8,822	687

Notes: Panel A counts are at the individual respondent level, pooled across the twelve KNHANES waves with cotinine data. Panel B counts are at the couple level. Couples are linked using the family relationship variables *relat* (2008–2009) and *fam_rela* (2010–2021), retaining one married opposite-sex respondent–spouse pair per household. An additional 4,931 married adults from multi-couple households (predominantly coresident elderly relatives in multi-generation households, mean age 61) are not linked under this rule and are excluded; see [section 3](#).

4 Empirical Strategy

Our empirical approach combines descriptive comparisons with regression-based tests of patterns motivated by the marriage-matching literature. I do not make any causal claims about concealment behaviour or its effects on marriage outcomes. The contributions are (i) to measure a gendered disclosure gap using biomarker validation, (ii) to document how this gap is correlated with spouse characteristics, and (iii) to show that empirical estimates of marital sorting assortativeness on smoking and education depend on whether smoking is measured by self-report or biomarker.

I organize the empirical analysis around five questions, each with its own specification and identifying content. Throughout, regressions include husband age, wife age, region fixed effects,

and survey wave fixed effects unless noted. Standard errors are heteroskedasticity-robust. I follow [Chen and Roth \(2024\)](#) in using the inverse hyperbolic sine transform when cotinine concentration enters the right-hand side of a regression.

4.1 Is there a gendered concealment gap?

I define the concealment indicator as following form:

$$\text{Conceal}_i = \mathbf{1}\{\text{cotinine}_i \geq 50\} \cdot \mathbf{1}\{\text{self-report nonsmoker}_i\},$$

which equals one for biomarker-positive individuals who self-report as nonsmokers and zero for biomarker-positive individuals who self-report as smokers. Within the biomarker-positive sample of adults (men and women, married and single, pooled), I estimate

$$\text{Conceal}_i = \alpha + \beta \text{Female}_i + \gamma' \mathbf{X}_i + \delta_r + \delta_t + \varepsilon_i, \quad (1)$$

where $\mathbf{X}_i$ includes age and indicators for marital status, δ_r and δ_t are region and wave fixed effects. The parameter β measures the gender difference in the probability of concealing biological smoking. I also estimate a specification with a female-by-married interaction to assess whether marital status amplifies the gender effect on concealment.

The identifying assumption is that, after conditioning on age, region, and wave, the comparison of self-reporting behaviour between men and women who are both biologically smoking recovers a gender difference in willingness to disclose. The principal threat is that men and women may differ systematically in smoking intensity (and hence in cotinine level near the 50 ng/mL cut-off), which could mechanically produce a disclosure gap even absent any difference in stigma response. I address this concern in two ways: by reporting the cotinine distributions of self-reported smokers and nonsmokers separately by sex ([Figure 1](#)), and by examining whether the disclosure gap survives intensity-based sample restrictions in [Section 4.6](#).

4.2 How does concealment relate to spouse characteristics?

Within the linked couples sample, I partition wives with valid cotinine into the three groups defined in [Section 3](#) (true nonsmoker, concealer, discloser). I then estimate

$$y_i = \alpha + \beta_C \text{Concealer}_i + \beta_D \text{Discloser}_i + \gamma' \mathbf{X}_i + \delta_r + \delta_t + \varepsilon_i, \quad (2)$$

where y_i is a husband characteristic or matching outcome (husband's education years, husband's high-education indicator, education gap, marry-up indicator (=1 if husband's years of schooling > wife's years of schooling), husband's self-reported smoking status, and husband's drinking status as a placebo); Concealer_i and Discloser_i are indicators with the true-nonsmoker group as the omitted baseline; and $\mathbf{X}_i$ includes wife age, wife education years (in the controlled specification), and husband age.

This regression does not identify a causal effect of concealment on marriage outcomes. Smoking and disclosure are observed only at the survey date, not at marriage formation, and I cannot determine whether the husband knows the wife's true smoking status. The coefficients β_C and β_D should therefore be interpreted as conditional correlations: how do the husband characteristics of concealing wives, and of disclosing wives, differ from those of true-nonsmoker wives with comparable observable characteristics? Under the stigma hypothesis I expect $|\beta_C| < |\beta_D|$ on outcomes for which smoking is socially penalized such that concealers should look more like true nonsmokers, disclosers less so. The husband-drinking regression serves as a placebo: if the stigma mechanism is smoking-specific, neither β_C nor β_D should differ from zero in that specification.

4.3 Concealment as outcome: an equivalent reverse framing

The three-group specification in equation (2) regresses husband characteristics on wife smoking type, with true nonsmokers as the omitted baseline. A mathematically related question asks which biologically smoking wives are more likely to conceal. I estimate

$$\text{Concealer}_i = \alpha + \delta \text{Edu}_i^H + \gamma' \mathbf{X}_i + \delta_r + \delta_t + \varepsilon_i, \quad (3)$$

on the subsample of 835 biomarker-positive wives, where Concealer_i equals one if the wife self-reports as a nonsmoker (and zero if she discloses as a smoker). The coefficient δ measures the conditional correlation between husband education and the wife's choice to disclose, holding fixed her own age and (in some specifications) her own education. I also estimate the specification with the husband high-education indicator, the spousal education gap, and the wife-marries-up indicator on the right-hand side.

This regression identifies the underlying correlation between concealment behaviour and husband characteristics, which is same as the equation (2) but from opposite directions. Moreover, as I restrict the sample in this regression on only biomarker-positive wives, I could compare the difference of husband characteristics between the concealers and disclosers. I expect to find the results in this section are consistent with what I can find in equation (2). With the results in this and

last section, I could set up the full ordering of husband characteristics between true-nonsmokers, concealers, and disclosers.

4.4 Self-report versus biomarker matching estimates

To quantify how the choice of measurement affects empirical estimates of marital sorting, I estimate three matching specifications under each of the two measurements. For smoking homogeneity:

$$\text{Smoker}_i^H = \alpha + \beta \text{Smoker}_i^W + \gamma' \mathbf{X}_i + \delta_r + \delta_t + \varepsilon_i, \quad (4)$$

estimated once with both smoking variables defined by self-report and once with both defined by cotinine. For the wife-smoking education gradient:

$$\text{Edu}_i^H = \alpha + \beta \text{Smoker}_i^W + \pi \text{Edu}_i^W + \gamma' \mathbf{X}_i + \delta_r + \delta_t + \varepsilon_i, \quad (5)$$

again estimated with both definitions of Smoker_i^W . I also estimate the symmetric specification with wife's education on the left-hand side and husband's smoking on the right.

The comparison across measurements addresses the central methodological question of the paper: do empirical conclusions about marital sorting on smoking depend on how smoking is measured? Under the null hypothesis that self-report and biomarker measurement are equivalent (i.e., that self-reported smoking measures biological smoking with classical error), the coefficient on Smoker_i^W should be similar across the two specifications. Systematic differences identify the magnitude of measurement-induced bias in matching estimates.

4.5 Time trend in under-reporting

I test whether the gender concealment gap has changed over time using two specifications. The linear trend specification is

$$\text{Conceal}_i = \alpha + \beta \text{Year}_t + \gamma' \mathbf{X}_i + \delta_r + \varepsilon_i, \quad (6)$$

estimated separately for biomarker-positive wives and (as a placebo) biomarker-positive husbands. The joint cohort-and-period specification is

$$\text{Conceal}_i = \alpha + \delta_c^{\text{cohort}} + \delta_t^{\text{wave}} + \gamma' \mathbf{X}_i + \delta_r + \varepsilon_i, \quad (7)$$

which includes both birth-cohort fixed effects and wave fixed effects. If both birth-cohort and wave fixed effects are significant, it suggests that both within-cohort norm change (period effect) and cohort replacement contribute to the trend variation.

4.6 Heterogeneity and robustness

Heterogeneity by stigma proxies. I test whether the concealment–husband-education correlation varies by birth cohort and by urban–rural location. Birth cohort proxies for the strength of traditional gender norms (older cohorts might face stronger gendered stigma); metropolitan residence proxies for exposure to more progressive or equal social environments. I estimate

$$\text{Conceal}_i = \alpha + \delta_1 \text{Edu}_i^H + \delta_2 Z_i + \delta_3 (\text{Edu}_i^H \times Z_i) + \gamma' \mathbf{X}_i + \delta_r + \delta_t + \varepsilon_i$$

where Z_i is the heterogeneity variable.

Smoking intensity robustness. A key threat to interpreting the concealment indicator as evidence of strategic disclosure is that “concealers” may simply be lighter smokers whose biomarker exposure is harder to detect socially. I address this concern in three ways. First, I examine whether cotinine levels differ between concealers and disclosers within the biomarker-positive sample. Second, I re-estimate the main three-group regression including the inverse hyperbolic sine of cotinine concentration as a control. Third, I restrict the sample to concealers and disclosers with cotinine values in the range 50–300 ng/mL which is lower than the 1000 ng/ml standard for frequent smokers, focusing on individuals with comparable biomarker intensity.

Male concealer placebo. As a symmetric test of whether the matching correlations are specific to female concealment, I estimate the three-group regression replacing the wife’s smoking type with the husband’s, and the husband characteristics on the left-hand side with the corresponding wife characteristics. Under the gendered-stigma interpretation, husband concealment should not predict wife characteristics in the way that wife concealment predicts husband characteristics, because male smoking is much less stigmatized in the Korean context.

Choo-Siow matching surplus. As a structural complement to the reduced-form matching comparison, I estimate marital surplus under each measurement of smoking following [Choo and Siow \(2006\)](#). I define four types per sex (low/high education $\times$ nonsmoker/smoker) and estimate

$$\hat{\Phi}_{xy} = 2 \log(\hat{\mu}_{xy}) - \log(\hat{\mu}_{x0}) - \log(\hat{\mu}_{0y}),$$

where $\hat{\mu}_{xy}$ is the observed count of couples in cell (x, y) and $\hat{\mu}_{x0}$, $\hat{\mu}_{0y}$ are counts of singles of each

type. I estimate the surplus matrix separately under self-report and cotinine classification and compare them. I do not report standard errors for the Choo-Siow estimates and treat the comparison as suggestive findings.

5 Results

I organize the results around the five empirical questions defined in Section 4. I first establish the existence and magnitude of the gendered concealment gap (Section 5.1). I then compare husband characteristics across the three smoking groups defined among married women (Section 5.2), and quantify how matching estimates differ between self-report and biomarker measurement (Section 5.4). Section 5.5 documents the decline in concealment over time, and Section 5.6 reports a series of robustness analyses.

5.1 The gender concealment gap

Figure 1a and Figure 1b plot the distribution of urinary cotinine separately by self-reported smoking status, for men and women respectively. Both distributions are sharply bimodal: cotinine values cluster near zero for self-reported nonsmokers and at the high end (typically 1,000 ng/mL or more) for self-reported smokers. Importantly, however, a substantial mass of self-reported nonsmokers among women carries cotinine values above the 50 ng/mL cutoff—visible as the right tail of the blue histogram in Figure 1b. The same overlap exists for men (Figure 1a) but is much smaller. This visual evidence motivates the regression analysis.

Table 3 reports under-reporting rates by sex and marital status, defined as the share of biomarker-positive adults who self-report as nonsmokers. The rates differ enormously by sex: 10.70% of biomarker-positive married men and 13.84% of biomarker-positive single men under-report, compared to 45.76% of biomarker-positive married women and 39.51% of biomarker-positive single women. Women under-report at roughly four to five times the male rate, with married women under-reporting somewhat more than single women.

Table 4 estimates equation (1) on the pooled biomarker-positive sample. Column (1) reports a basic specification with wave and region fixed effects: women are 33.9 percentage points more likely than men to self-report as nonsmokers, conditional on biomarker-positive status. The coefficient is statistically significant. Column (2) adds age as control variable and the magnitude is essentially unchanged. Column (3) adds a female-by-married interaction: the female main effect is 25.9 percentage points (for single women), and married women face an additional 9.8 percentage point increment, which implied married women are more likely than single women to

Table 3: Under-reporting rates among biomarker-positive adults

Group	#(cotinine+)	#(under-report)	Rate (%)
Male, married	8050	861	10.70
Male, single	2623	363	13.84
Female, married	2275	1041	45.76
Female, single	615	243	39.51

Notes: Under-reporter = biomarker-positive but self-reports as nonsmoker. Smoker definition: currently smoking (daily or occasional) with lifetime $\geq$ 100 cigarettes.

self-report as nonsmokers even though they are cotinine-positive smokers. Both terms are highly significant. Column (4) replicates the specification using the stricter daily-only smoker definition; the female main effect remains 26.0 percentage points and the marriage interaction increases to 12.3 percentage points.

Table 4: Gender concealment gap among biomarker-positive adults

	(1)	(2)	(3)	(4)
Female	0.339*** (0.010)	0.337*** (0.010)	0.259*** (0.021)	0.260*** (0.022)
Age		-0.001*** (0.000)	-0.002*** (0.000)	-0.003*** (0.000)
Married			0.005 (0.009)	0.010 (0.010)
Female $\times$ Married			0.098*** (0.024)	0.123*** (0.024)
Wave FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Daily Smoker Status	No	No	No	Yes
Observations	13232	13232	13232	13563

Notes: Standard errors in parentheses. Column 4 uses daily-only smoker definition (robustness). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

These magnitudes are large. A 34 percentage point gender gap in disclosure of a biologically-verified behaviour is larger than any previously documented gender concealment gap. [Exley et al. \(2024\)](#) find a smaller but still substantial gender concealment gap in academic grades but in the opposite direction (male students conceal more in their context). The direction and magnitude of the concealment gap thus depends on which gender faces social penalty for the trait being concealed.

5.2 Three-group comparison: how does concealment relate to spouse characteristics?

Next, I explore whether biomarker-positive concealers and disclosers differ systematically from biomarker-negative true nonsmokers in their husband characteristics. Among the 8,768 married women in our analytic sample, 7,933 are true nonsmokers, 473 are concealers, and 362 are disclosers.

[Table A.2](#) reports descriptive means of wife and husband characteristics across the three groups. In raw means, concealers are younger than true nonsmokers (mean age 45.8 versus 51.7) and similar to disclosers (44.9). Concealers also have higher mean education (9.80 years) than both true nonsmokers (8.89) and disclosers (9.21). However, this unconditional educational ordering largely reflects the age difference: younger Korean women are more educated overall.

[Table A.3](#) reports regressions of wife's own characteristics on the three-group categorical variable, controlling for age, region, and wave. Conditional on these controls, the educational ordering reverses. Concealers have 0.65 fewer years of schooling than comparable true nonsmokers ($p < 0.01$), and disclosers have 1.36 fewer years. The high-education indicator follows the same pattern: concealers are statistically indistinguishable from true nonsmokers on this binary measure, while disclosers are 12.4 percentage points less likely to have completed high school. Both concealers and disclosers are substantially more likely than true nonsmokers to drink monthly (+10.6 and +21.3 percentage points respectively), consistent with smoking and drinking being correlated health behaviours.

[Table 5](#) reports the central matching result: regressions of husband characteristics on the three-group categorical, controlling for wife age, wife education, husband age, region, and wave (equation (2)). Column (1) examines husband education. Concealers' husbands have 0.37 fewer years of schooling than true nonsmokers' husbands ($p < 0.01$); disclosers' husbands have 0.75 fewer years ($p < 0.01$). The concealer penalty is approximately half the discloser penalty. Column (2) shows that the high-education husband indicator follows a similar pattern: concealers are statistically indistinguishable from true nonsmokers on this margin, while disclosers are 4.4 percentage points less likely to have a high-education husband. The education gap (Column 3) and marry-up indicator (Column 4) shows the same patterns.

Columns (5) and (6) report regressions on husband smoking status, measured by self-report and biomarker respectively. Both concealers and disclosers are roughly 21–26 percentage points more likely than true nonsmokers to have a smoking husband. The two coefficients (concealer and discloser) are similar within each measurement, suggesting that the partner-smoking dimen-

Table 5: Husband characteristics by wife smoking type

	(1) Husband Education	(2) Husband High-edu	(3) Education Gap	(4) Marry up	(5) Husband Smoke SR	(6) Husband Smoke COT	(7) Husband Drink
Concealer	-0.367*** (0.138)	0.008 (0.016)	-0.367*** (0.138)	-0.037 (0.023)	0.211*** (0.022)	0.232*** (0.020)	0.002 (0.021)
Discloser	-0.753*** (0.175)	-0.044** (0.021)	-0.753*** (0.175)	-0.060** (0.025)	0.256*** (0.024)	0.222*** (0.024)	0.015 (0.023)
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wife Edu	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8768	8768	8768	8768	8726	8768	8739

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

sion of matching operates on biological status rather than on disclosure. Column (7) reports the placebo regression on husband monthly drinking: the coefficients on both concealer and discloser are small (0.002 and 0.015) and not statistically significant. The placebo holds: wife smoking type predicts husband smoking and husband education, but not husband drinking, consistent with stigma being smoking-specific.

The three-group pattern is informative about the interpretation of concealment. Concealers' husbands are systematically less educated than the husbands of comparable true nonsmokers, but the gap is roughly half that of disclosers. The pattern is consistent with concealment being correlated with marriage-market outcomes that partially recover the disclosed-smoker penalty.

5.3 Concealment as outcome: who conceals?

An alternative framing of the same correlation asks which biomarker-positive wives are most likely to conceal. [Table 6](#) reports regressions of the concealer indicator on husband characteristics, within the subsample of 835 biomarker-positive wives.

Column (1) shows that each additional year of husband education is associated with a 1.6 percentage point higher probability of wife concealment ($p < 0.01$). Column (2) reports the same relationship using the binary high-education indicator: wives married to high-education husbands are 14.5 percentage points more likely to conceal their smoking ($p < 0.01$). Columns (3) and (4) examine the education gap and marry-up indicator; neither is significantly associated with concealment.

Column (5) re-estimates the husband-education specification while controlling for the wife's own education. The husband-education coefficient declines from 0.016 to 0.013 but remains significant at the 5% level. The wife's own education coefficient (0.008) is not significant. Concealment is therefore associated with husband education even conditional on the wife's own education, though I cannot determine whether this conditional association reflects husband-specific stigma response, the wife's marital incentive to protect the marriage duration, or unobserved characteristics correlated with both.

This regression is descriptively equivalent to the three-group comparison in Section 5.2 and both characterize the same correlation between concealment behaviour and husband characteristics. The +0.016 coefficient in column (1) is consistent with the three-group comparison in Section 5.2: the implied concealer-vs-discloser contrast from the three-group regression ($\beta_C - \beta_D = +0.386$ years of husband education) matches the within-biomarker-positive correlation identified here. Both regressions recover the same fact from different perspectives: among biological smokers, wives married to higher-educated husbands are more likely to conceal.

Table 6: Concealment as outcome

	(1)	(2)	(3)	(4)	(5)
Husband Education	0.016*** (0.005)				0.013** (0.006)
Husband Hige-edu		0.145*** (0.045)			
Education Gap			0.004 (0.005)		
Marry Up				0.018 (0.036)	
Wife Education					0.008 (0.006)
Wave FE	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
Age Control	Yes	Yes	Yes	Yes	Yes
Observations	835	835	835	835	835

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.4 Self-report versus biomarker matching estimates

The central methodological contribution of the paper is to quantify how matching estimates differ when smoking is measured by self-report versus biomarker. Table 7 reports three matching specifications, each estimated twice (once with self-reported smoking, once with cotinine-defined smoking).

Table 7: Self-report vs biomarker matching coefficients

	(1) Husband Smoking SR	(2) Husband Smoking COT	(3) Husband Education	(4) Husband Education	(5) Wife Education	(6) Wife Education
Wife Smoking (self-reported)	0.240*** (0.024)		-0.744*** (0.170)			
Wife Smoking (biomarker)		0.235*** (0.017)		-0.535*** (0.114)		
Wife Education			0.624*** (0.010)	0.624*** (0.010)		
Husband Smoking (self-reported)					-0.201*** (0.064)	
Husband Smoking (biomarker)						-0.212*** (0.064)
Husband Education					0.530*** (0.009)	0.529*** (0.009)
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Age Control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8743	8822	8785	8822	8777	8822

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Columns (1) and (2) report the smoking homogamy regression—husband’s smoking status on wife’s smoking status. Under self-report, the coefficient is 0.240; under cotinine measurement, it is 0.235. The two estimates are all statistically significant with similar magnitude. Smoking homogamy estimates are robust to measurement choice in this setting.

The contrast appears in the gradient specifications. Columns (3) and (4) report husband education on wife smoking, controlling for wife education. Under self-report, smoking wives have husbands with 0.74 fewer years of schooling than nonsmoking wives ($p < 0.01$). Under cotinine measurement, the same gradient is 0.54 fewer years. Self-report overstates the wife-smoking edu-

cation penalty by approximately 39%.

Columns (5) and (6) report the symmetric regression: wife education on husband smoking, controlling for husband education. The estimates are all significant and similar across measurements (-0.20 under self-report, -0.21 under cotinine). For the husband-smoking direction, measurement choice does not affect the gradient.

The asymmetry between the wife-direction and husband-direction results is meaningful. Female concealment is large (45.8% of biomarker-positive married women under-report), so self-reported female smoking identifies a small, negatively selected subset of biological female smokers. The wife-smoking education gradient estimated on this selected sample overstates the population gradient by nearly 39%. Male concealment is small (10.7% of biomarker-positive married men), so self-reported male smoking is much closer to biological male smoking, and the husband-smoking estimates are robust across measurements.

The homogamy estimates remain robust to measurement choice because both spouses' smoking variables are measured consistently (both self-report, or both biomarker). The gradient estimates, in contrast, compare a smoker subgroup against shared partners and depend on which wives are classified as smokers. When the smoking-wife group changes systematically across measurement schemes, the gradient changes accordingly.

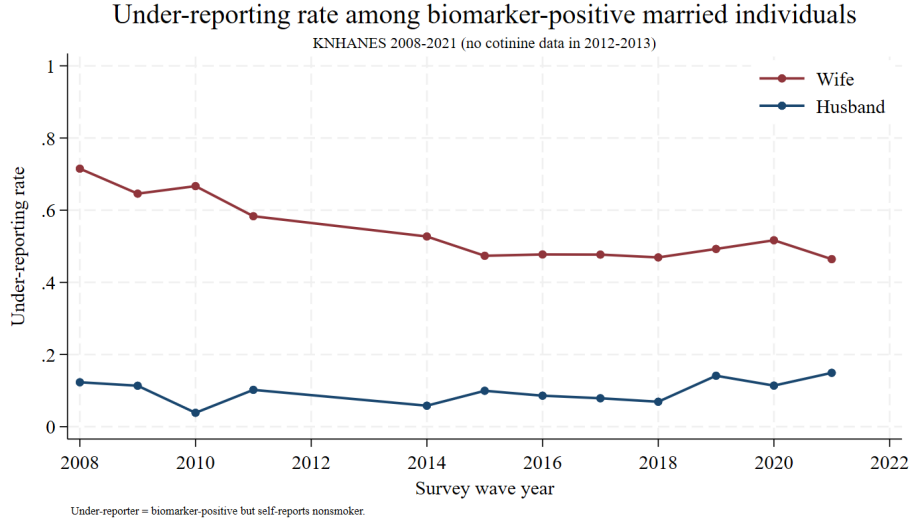
5.5 Time trend and cohort decomposition

Figure 2 plots the under-reporting rate by survey wave, separately for biomarker-positive wives and husbands. Female under-reporting falls from approximately 71% in 2008 to 46% in 2021, a decline of roughly 25 percentage points over thirteen years. Husband under-reporting fluctuates around 10% throughout the period with no clear trend, serving as a placebo against secular changes in survey response behaviour.

Table A.4 estimates equation (6) for the wife and husband samples separately. Column (1) shows a wife linear time trend of -2.2 percentage points per year (robust SE 0.4), implying a 28.6 percentage point decline over the 2008–2021 period. The coefficient is highly significant ($p < 0.01$). Column (2) replicates the specification for husbands: the trend is -0.3 percentage points per year, which implies the concealment is also declining for smoking husband but the magnitude is trivial compared with wife's trend.

Column (3) reports the joint cohort-and-period specification (equation (7)) among biomarker-positive wives. The post-1980 birth cohort dummy is -10.5 percentage points (relative to the

Figure 2: Trend of Under-reporting Rate of Smoking Status by Gender, 2008–2021



omitted 1960s cohort) and is statistically significant, indicating that the youngest cohort under-reports substantially less even conditional on wave-level controls. Earlier-born cohorts (pre-1950 and 1950s) trend slightly higher than the 1960s baseline, though these coefficients are not consistently significant. The joint specification suggests that both period effects (within-cohort norm change) and cohort replacement contribute to the aggregate decline, with the post-1980 cohort representing a discrete break from previous cohort baseline.

5.6 Robustness

In this section, I report four robustness analyses: smoking intensity, male concealer placebo, regional heterogeneity, and structural matching surplus.

Smoking intensity. A natural concern is that biomarker-positive concealers are simply lighter smokers than disclosers, with cotinine concentrations near the 50 ng/mL cutoff. [Table A.5](#) addresses this in three ways. Column (1) regresses the asinh-transformed cotinine concentration on the concealer indicator (within the biomarker-positive wife sample). The coefficient is -0.75 ($p < 0.01$), confirming that concealers do have lower cotinine concentrations than disclosers. However, the descriptive statistics in Section 5.2 ([Table A.2](#)) showed that concealers' average cotinine is approximately 639.5 ng/mL—well above the 50 ng/mL cutoff and clearly in the smoking range. This result suggests that concealers are lighter smokers, but not borderline just around the cutoff.

Column (2) re-estimates the husband education regression including the asinh cotinine concentration as a control. The concealer-versus-discloser coefficient on husband education is +0.46 years ($p < 0.05$), and the cotinine coefficient itself is small and insignificant. The concealer-discloser gap in husband education survives intensity control. Column (3) restricts the sample to wives with cotinine concentrations between 50 and 300 ng/mL—a narrower intensity range. The sample size shrinks to 251 and the coefficient is no longer significant, though the point estimate (−0.19) goes in the opposite direction. On the one hand, I interpret the loss of significance as a power limitation rather than evidence against the main result; on the other hand, this enhances our main specification results as it indicates that the structural differences are not driven by variations across groups which are around the cutoff.

Male concealer placebo. Table A.6 reports the symmetric specification with husband concealment predicting wife characteristics. The sample consists of 3,707 biomarker-positive husbands (with *h_concealer* defined). Four of five outcomes show coefficients statistically indistinguishable from zero: wife education (+0.18, $p = 0.23$), wife high-education indicator (+0.012, $p = 0.45$), education gap (−0.18, $p = 0.23$), and wife marry-up indicator (+0.004, $p = 0.82$). The fifth outcome, wife self-reported smoking, shows a small but statistically significant negative coefficient (−0.061, $p < 0.01$). This last result reflects biomarker-level smoking homogamy operating in the opposite direction (concealing husbands are biological smokers and smoking is positively correlated between spouses on the biomarker dimension), not a matching effect.

The male placebo result supports our interpretation that the female concealment–husband education correlation reflects gendered stigma. If the correlation arose from a more general mechanism unrelated to gender-specific smoking norms, I would expect husband concealment to predict wife education in a comparable way, while I do not see any evidence support the existence of other general mechanism.

Regional and cohort heterogeneity. Table A.7 reports heterogeneity analyses. Column (1) shows the cohort gradient in wife under-reporting: pre-1950 cohort +14.3 percentage points relative to the 1960s baseline (marginally significant with $p < 0.1$), 1980+ cohort −14.4 percentage points (highly significant with $p < 0.01$). Column (2) reports the metropolitan-province comparison: the metropolitan dummy coefficient is −0.021 ($p = 0.55$). I interpret this as a meaningful null result. Female smoking stigma is uniform across Korean geography; it is not concentrated in rural or traditional regions.

Column (3) reports the concealment-husband education interaction by cohort. The main effect of husband education (for the omitted 1960s cohort) is +3.5 percentage points per year

($p < 0.01$): a wife in this cohort with a husband one year more educated is 3.5 pp more likely to conceal. The interaction terms with other cohorts are predominantly negative: -4.2 pp for the pre-1950 cohort, -3.2 pp for the 1950s cohort, -2.6 pp for the 1970s cohort (all significant). The 1980+ interaction is small and insignificant. The concealment–husband-education correlation is strongest for the 1960s birth cohort and weaker on either side, suggesting that this cohort having experienced a unique combination of strong residual stigma and active marriage-market opportunities for upward education mobility.

Choo-Siow matching surplus. As a structural complement to the reduced-form matching comparison, I estimate the Choo-Siow joint marital surplus $\hat{\Phi}_{xy}$ under each measurement of smoking. Table A.8 reports surplus estimates on the four-by-four type matrix (low/high education $\times$ nonsmoker/smoker per sex), focusing on the four diagonal cells corresponding to same-type homogamy. Both nonsmoker diagonals (low-education nonsmoker–low-education nonsmoker, and high-education nonsmoker–high-education nonsmoker) show higher surplus under self-report than under cotinine measurement: $\hat{\Phi}_{1,1}^{SR} = 4.78$ versus $\hat{\Phi}_{1,1}^{COT} = 4.43$, and $\hat{\Phi}_{3,3}^{SR} = 1.02$ versus $\hat{\Phi}_{3,3}^{COT} = 0.42$. Both smoker diagonals show the opposite pattern: surplus is higher under cotinine than under self-report.

This sign pattern is informative. Under self-report, biomarker-positive concealers are incorrectly classified as nonsmokers, upward biasing the apparent surplus of both-nonsmoker matches and deflating the apparent surplus of both-smoker matches. The cotinine measurement, by correctly classifying these wives, produces a flatter matching surplus distribution across the type space. Self-report systematically distorts the inferred matching surplus pattern in the direction predicted by concealment behaviour. I treat the comparison as exploratory evidence. The qualitative pattern is consistent with the reduced-form measurement comparison in Section 5.4: measurement choice changes empirical conclusions about marital sorting on smoking.

6 Discussion and Concluding Remarks

Social norms and stigma are difficult to observe directly, yet they shape economically important human decisions. This paper shows that when a stigmatized behaviour is observed both through self-report and through an objective biomarker, the gap between the two measures can be used to identify concealment behaviour. I apply this idea to female smoking in South Korea, where smoking by women, especially married women, is subject to strong social stigma. By comparing self-reported smoking status with urinary cotinine, I separate biological smoking from willingness to disclose smoking.

The results document a large and systematic gendered concealment gap. Among biomarker-positive adults, women are substantially more likely than men to report themselves as nonsmokers, and the gap is larger among married women. This pattern is consistent with the interpretation that concealment is a behavioural response to gendered stigma rather than random reporting error. The analysis also shows that concealment is not random within the marriage market. Among biomarker-positive wives, those married to higher-educated husbands are more likely to conceal, and concealers occupy an intermediate position between true nonsmokers and disclosers in terms of husband characteristics. These patterns suggest that disclosure behaviour is related to partner characteristics and social incentives within marriage.

The paper also shows that stigma-induced misreporting has consequences for empirical estimates. When smoking is measured by self-report, the estimated relationship between wife smoking and husband education is substantially larger in magnitude than when smoking is measured by biomarker. This occurs because self-reported female smokers are a selected subset of biological smokers: they are not only smokers, but smokers who are willing to disclose. In contrast, smoking homogamy estimates become less when correcting the measurement with biomarker-validated smoking status. These findings illustrate a broader measurement problem for empirical work on marriage matching that self-reported measures may distort estimated sorting patterns when a matching trait is socially stigmatized and disclosure differs systematically across groups.

The time-series evidence further suggests that the social cost of female smoking has weakened over time. Concealment among biomarker-positive wives declines markedly between 2008 and 2021, while concealment among husbands remains low and relatively stable. The cohort analysis indicates that both within-cohort changes and cohort replacement contribute to this decline, with women born after 1980 showing lower concealment. This pattern is consistent with gradual changes in gender norms and the declining stigma attached to female smoking in Korea.

Several limitations should be noted. The analysis does not identify the causal effect of concealment on marriage outcomes. Smoking status and disclosure are observed at the survey date, not at the time of marriage formation, and I cannot observe whether husbands know their wives' true smoking status. The matching results should therefore be interpreted as conditional correlations between concealment behaviour and spouse characteristics, rather than causal effects of concealment on partner choice. In addition, urinary cotinine measures recent nicotine exposure rather than lifetime smoking preferences, and some misclassification may arise from occasional smoking, second-hand smoke exposure, or variation in smoking intensity. The results are therefore best interpreted as evidence on biomarker-validated concealment of recent smoking, not as a complete measure of all smoking behaviour.

Despite these limitations, the findings have two broader implications. First, they show that biomarker data can be used not only to correct measurement error, but also to study stigma itself through the behaviours it induces. The object measured here is not stigma directly, but stigma-induced concealment. Second, the results suggest that researchers should be cautious when using self-reported measures of socially sensitive traits, especially in settings where stigma is gendered or group-specific. Similar concerns may arise for other traits relevant to marriage, labour markets, and social behaviour, including drug use, sexual behaviour, mental health, and politically sensitive attitudes. Where objective validation is available, comparing reported and validated status can reveal both the extent of mismeasurement and the social forces that generate it.

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Online Appendix – Not for Print
Stigma, Concealment, and Marriage Matching
Ye Yuan

A Appendix Figures and Tables

Table A.1: Summary statistics, full sample (appendix)

	Married men	Married women	Single men	Single women
<i>Panel A: Demographics</i>				
Age	55.596 (14.129)	54.822 (14.840)	29.216 (9.831)	27.236 (9.371)
Education (years)	9.704 (4.819)	7.825 (5.268)	12.420 (2.822)	12.947 (2.501)
High education (HS+)	0.677	0.558	0.912	0.940
Body mass index	24.293 (3.134)	23.744 (3.500)	24.168 (4.093)	21.819 (3.798)
<i>Panel B: Smoking (among those with cotinine)</i>				
Lifetime ≥ 100 cigarettes	0.796	0.091	0.579	0.148
Self-report current smoker	0.356	0.046	0.409	0.090
Self-report daily smoker	0.329	0.038	0.366	0.071
Cotinine-positive (≥ 50 ng/mL)	0.474	0.116	0.548	0.195
Cotinine level (ng/mL)	630.266 (894.315)	100.855 (358.908)	720.825 (922.282)	153.489 (415.051)
<i>Panel C: Other behavior & location</i>				
Monthly drinker	0.702	0.358	0.724	0.576
Metropolitan resident	0.429	0.433	0.494	0.537
Observations (all)	27550	37969	7283	6406
Observations (cotinine)	16989	19640	4787	3153

Notes: Full sample of KNHANES 2008–2021 adult respondents (pooled across waves). Panel B smoking variables computed among those with valid urinary cotinine measurement; denominator differs from Panel A and C variables, which use the full sample.

Table A.2: Characteristics by wife smoking type

	True nonsmoker	Concealer	Discloser
<i>Wife characteristics</i>			
Age	51.653	45.797	44.870
Education (years)	8.892	9.801	9.210
High education	0.617	0.732	0.677
Monthly drinker	0.375	0.529	0.657
Cotinine level	3.600	639.498	1027.414
<i>Husband characteristics</i>			
Age	54.935	48.717	47.564
Education (years)	10.192	10.579	9.710
High education	0.692	0.755	0.713
Self-report smoker	0.357	0.672	0.743
Cotinine smoker	0.388	0.744	0.762
Monthly drinker	0.720	0.761	0.785
Observations	7933	473	362

Notes: Sample of married couples with cotinine on both spouses. True nonsmoker: cotinine-negative and self-reports nonsmoker. Concealer: cotinine-positive but self-reports nonsmoker. Discloser: cotinine-positive and self-reports smoker.

Table A.3: Wife's own characteristics by smoking type

	(1) Edu years	(2) High-edu	(3) Drinker
Concealer	-0.649*** (0.154)	-0.009 (0.017)	0.106*** (0.023)
Discloser	-1.363*** (0.178)	-0.124*** (0.022)	0.213*** (0.025)
Wave FE	Yes	Yes	Yes
Region FE	Yes	Yes	Yes
Age Control	Yes	Yes	Yes
Observations	8768	8768	8766

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.4: Time trend in under-reporting

	(1) Wife trend	(2) Husband trend	(3) Cohort+Period
Wave Year	-0.022*** (0.004)	-0.003** (0.001)	
Wife Age	0.006*** (0.002)		
Wife High-edu	0.185*** (0.046)		0.169*** (0.048)
Husband Age		0.001** (0.000)	
Husband High-edu		0.020 (0.012)	
Pre-1950 Birth Cohort			0.104 (0.076)
1950s Birth Cohort			0.079 (0.058)
1960s Birth Cohort			0.000 (.)
1970s Birth Cohort			-0.033 (0.049)
Post-1980 Birth Cohort			-0.105** (0.053)
Wave FE	No	No	Yes
Region FE	Yes	Yes	Yes
Observations	848	3728	848

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.5: Smoking intensity robustness

	(1) Wife Cotinine (asinh)	(2) Husband Education	(3) Restricted 50-300 ng/ml
Wife Concealer	-0.750*** (0.069)	0.456** (0.227)	-0.189 (0.587)
Wife Cotinine (asinh)		-0.008 (0.101)	
Wave FE	Yes	Yes	Yes
Region FE	Yes	Yes	Yes
Age Control	Yes	Yes	Yes
Observations	835	835	251

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.6: Male concealer placebo

	(1) Wife Education	(2) Wife High-edu	(3) Education Gap	(4) Wife Marry-up	(5) Wife Smoking SR
Husband Concealer	0.179 (0.150)	0.012 (0.016)	-0.179 (0.150)	0.004 (0.019)	-0.061*** (0.009)
Wave FE	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
Age Control	Yes	Yes	Yes	Yes	Yes
Observations	3707	3707	3707	3707	3694

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.7: Heterogeneity by stigma proxies

	(1)	(2)	(3)
Pre-1950 Birth Cohort	0.143* (0.075)		0.314 (0.197)
1950s Birth Cohort	0.091 (0.058)		0.299** (0.143)
1960s Birth Cohort	— —		— —
1970s Birth Cohort	-0.040 (0.049)		0.328** (0.163)
Post-1980 Birth Cohort	-0.144*** (0.052)		0.152 (0.212)
Wife High-Edu	0.183*** (0.048)	0.164*** (0.045)	
Metro		-0.021 (0.035)	
Wife Age		0.005*** (0.002)	0.006 (0.005)
Husband Education			0.035*** (0.010)
Pre-1950 × Husband Edu			-0.042*** (0.016)
1950s × Husband Edu			-0.032** (0.013)
1960s × Husband Edu			— —
1970s × Husband Edu			-0.026** (0.013)
Post-1980 × Husband Edu			-0.014 (0.016)
Wife Education			0.005 (0.007)
<i>N</i>	848	848	835

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.8: Choo-Siow joint marital surplus estimates: self-report vs cotinine

Wife type	Husband type			
	LE, NS	LE, S	HE, NS	HE, S
<i>Panel A: Self-report measurement</i>				
LE, NS	4.78 [3,317]	3.40 [1,270]	0.96 [1,566]	-0.66 [580]
LE, S	-1.70 [53]	-0.49 [74]	-5.15 [30]	-3.81 [49]
HE, NS	-1.92 [495]	-2.21 [326]	1.02 [6,846]	0.31 [4,008]
HE, S	-6.30 [17]	-4.56 [31]	-5.28 [90]	-2.79 [261]
<i>Panel B: Cotinine measurement</i>				
LE, NS	4.43 [1,403]	2.79 [687]	0.48 [634]	-1.00 [333]
LE, S	-1.12 [56]	-0.01 [108]	-4.60 [32]	-3.89 [50]
HE, NS	-2.17 [221]	-3.12 [152]	0.42 [2,624]	-0.39 [1,920]
HE, S	-5.72 [18]	-4.28 [41]	-4.54 [106]	-1.89 [437]

Notes: Surplus estimates follow [Choo and Siow \(2006\)](#): $\Phi_{xy} = 2\log(\mu_{xy}) - \log(\mu_{x0}) - \log(\mu_{0y})$, where μ_{xy} is the number of couples of type (x, y) and μ_{x0}, μ_{0y} are single counts. Types defined by education (LE = less than high school graduate; HE = high school or above) and smoking status (NS = nonsmoker; S = smoker). Cell counts (couples) in brackets. Diagonal entries (same-type homogamy) shown in bold. Standard errors not computed; estimates are exploratory.